reference for other applications
- `tmp` – As the name suggests, it acts as storage for temporary files
- `usr` – This is where most of your applications and files will be stored, as anything present here is available for all users to access
- `var` – This is a directory for variable files such as logs and databases. Notice the contrast with the `/tmp` directory.

3.2 Converting an ext3 file system into ext4

Ext4 file system provides better performance and faster file system check than the ext3 file system which was in use until very recently. With Kernel 2.6.28, ext4 was marked stable, and with a subsequent Ubuntu 9.04 (Jaunty Jackalope) release, you can even use Ext4 for a fresh install. But since most of you are stuck with the older ext3 file system, we will see how to convert it into the new ext4 file system.

First thing that you should do is check your existing file system to be sure that you have an Ext4 aware kernel, else you might end up with a non-bootable brick. Type in the command

`uname –a` and make sure that its higher than `2.6.28`.

Next, you need to switch over to the Ext4 driver without changing the existing files on disk. This is made possible because of the fact that the Ext4 driver is backwards compatible and can mount an Ext3 file system. Type the following command:

```
sudo nano /etc/fstab
```

and look for the ext3 on the line that defines your disk(sda1 in most cases) and change it to `ext4`. If you have more than one partition for your Linux set-up, change for all of them and reboot your computer.

After this, you are ready to enable the new Ext4 features for your system with the following command:

```
sudo tune2fs -O extents,uninit_bg,dir_index <dev>
```

where dev stands for the disk edited in the previous step, `/dev/sda1` in our example. You need to again reboot your system for this to take effect and to run a file system check which can only be done on an unmounted file system.

You might see a lot of warnings of file system issues on your screen. It is perfectly normal and you've nothing to worry about. All that is left now is reinstalling GRUB. The GRUB boot loader that you will install will also, obviously have to be a version later than when ext4 was marked stable, so that it understands your file system and actually boots up your computer. Do a `sudo grub-install <dev>` to make sure that your grub version is the latest.

All the files written onto the disk after this will be able to take full advantage of the ext4, while your previous files will still be of ext3 format.